

How to Run a Tetrad Test in Your Facility:

Sample Preparation & Test Setup for Wine and Cider Tetrad Tests

This section outlines exact, step-by-step guidance for preparing samples and setting up a tetrad test so results are reliable and defensible for production decisions.

1. Clearly define the test objective in writing
 - a. This test method is effective for the following test objectives:
 - i. To determine whether a perceptible difference exists between samples
 - ii. To select, train, and monitor assessors
2. Before preparing samples, clearly define:
 - a. Product A: control/reference (e.g., current process, standard yeast)
 - b. Product B: test variation (e.g., new yeast, fining change, altered SO₂)
 - c. Best practices:
 - i. Only one variable should differ
 - ii. All other factors (vintage, lot, container, storage) must be identical
 - iii. Confirm both products are finished, stable, and representative
3. Decide Panel Size and Serving Orders
 - a. Recommended Panel Sizes:
 - i. 24 tasters → very common, efficient, balanced
 - ii. 30 tasters → stronger conclusions
 - iii. More tasters = increased sensitivity, but also more coordination

Why 24 or 30 Works Well:

These allow you to evenly distribute the six valid tetrad serving orders:

A A B B	B A B A
B B A A	A B B A
A B A B	B A A B

Each order should be used the same number of times.

4. Sample Volume & Glassware
 - a. Pour Volumes
 - i. Wine: 30 mL per sample
 - ii. Cider: 30 mL per sample
 - b. Each taster receives four samples, so plan accordingly.
 - c. Always prepare extra volume (10–15%) to account for spills and errors.
 - d. Glassware:
 - i. Use identical glasses for all samples
 - ii. Avoid brand-marked glassware
 - e. Consider:
 - i. Clear ISO-style glasses if color differences are acceptable
 - ii. Black or opaque glasses if color could bias judgments (common in cider)
 - f. Clean, odor-free, and dry glasses only
 - g. No sanitizer residue
5. Temperature Control
 - a. Serve samples at appropriate and consistent temperatures

- b. All four samples for a taster must be the same temperature
- 6. Pouring & Sample Coding
 - a. Sample Coding
 - i. Use random 3-digit codes (for example: 418, 702, 935)
 - b. Codes must:
 - i. Be unique per session
 - ii. Not repeat across Product A and B
 - iii. Never use letters, tank numbers, or lot names
 - c. Coding Tip
 - i. Prepare a worksheet (see additional documents) that links:
 - ii. Code → Product A or B
 - iii. Serving order → each taster
 - iv. Keep this key hidden until after analysis.
 - d. Pouring Procedure
 - i. Label glasses before pouring
 - ii. Pour all Product A samples first, then Product B (or vice versa)
 - iii. Rotate the pouring order between sessions to avoid subtle bias
 - iv. Cover glasses (coasters or lids) if serving is delayed
 - v. Avoid splashing, aeration differences, or inconsistent foam (especially for cider)
- 7. Sample Holding & Timing
 - a. Avoid holding poured samples longer than 10–15 minutes
 - b. If needed:
 - i. Cover glasses
 - ii. Hold at serving temperature
 - c. For carbonated cider:
 - i. Pour as close to serving time as possible
 - ii. Keep carbonation consistent across samples
 - iii. If samples evolve rapidly (oxidation, foam loss, temperature change), consider sequential serving instead of simultaneous.
- 8. Testing Environment
 - a. Room Setup
 - i. Quiet space, away from the cellar or production floor
 - ii. Minimal background odors
 - iii. Consistent lighting
 - b. At Each Tasting Station Provide:

- Four coded samples - Water - Spit cup - Plain crackers or bread	- Pencil and score sheet (see additional documents) - No phones - No talking - No coaching
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- 9. Test Procedure:

- a. Present each set of four samples simultaneously if possible, following the same spatial arrangement for each assessor.
 - b. Instruct the assessors to evaluate the four test samples in the order presented. The assessor should then group the four samples into two groups of two based on similarity.
 - c. Instructions to Tasters: “You will receive four samples. Two samples are the same, and two samples are different. Taste the samples in the order presented. Your task is to group the four samples into two pairs based on similarity. You may taste samples more than once. Even if you cannot tell a difference, please make your best guess. Do not choose based on preference.”
 - d. Do not say:
 - i. “Find the two that are most similar”
 - ii. “Identify the odd sample”
 - iii. “Choose the best one”
 - e. During the Test:
 - f. Allow tasters to proceed at their own pace
 - g. Do not answer questions about:
 - i. Product identity
 - ii. Production changes
 - iii. Expected outcomes
 - h. Collect score sheets before any discussion. If a taster finishes early, have them remain silent or exit quietly.
10. After the Test:
- a. Collect all score sheets
 - b. Do not discuss results until all data are recorded
 - c. Count how many correct groupings occurred
 - d. Analyze results based on your chosen panel size and goal
 - e. Communicate findings clearly:
 - i. Difference detected
 - ii. No difference detected (within test sensitivity)

Practical Tips from Winery/Cidery Use:

- Run a practice session with water or juice when training staff
- Test mid-morning or early afternoon (fatigue matters)
- Document everything — sample source, temperature, panel size, codes